Azure Backup & Site Recovery — Deployment & Configuration

Implementation Solution Guide

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Contents

Executive Summary — Azure Backup & Site Recovery — Deployment & Configuration 4

The problem . . . 4

The solution . . . 4

What you get . . . 5

Business outcomes . . . 5

Effort . . . 5

1. Solution Overview — Azure Backup & Site Recovery — Deployment & Configuration 5

Description . . . 5

How the technology works . . . 6

Deployment model — Assess → Configure → Pilot → Validate → Expand . . . 6

Named objects . . . 7

Scope . . . 7

Authentication and connections used . . . 8

2. Requirements — Azure Backup & Site Recovery — Deployment & Configuration 8

R1 — Agreed RPO/RTO targets per workload class (business decision) . . . 9

R2 — A secondary (target) Azure region for ASR . . . 9

R3 — Vault soft-delete and immutability policy decision . . . 9

R4 — RBAC roles for backup and Site Recovery . . . 10

R5 — Network line of sight for replication and (if applicable) the on-prem ASR appliance . . . 10

R6 — Application-consistency prerequisites for SQL-in-VM and app VMs . . . 10

R7 — An isolated DR test network and restore target (ties to GR2) . . . 11

R8 — Az PowerShell modules and Bicep tooling on the management host . . . 11

Global Requirements — Backup & DR Solutions 11

GR1 — A tenant/subscription with backup tooling and administrative access . . . 11

GR2 — A non-production restore target . . . 12

GR3 — Documented RPO/RTO objectives . . . 12

GR4 — Secure evidence storage . . . 12

GR5 — Change control . . . 13

GR6 — Network connectivity and a tooling host . . . 13

3. Implementation — Azure Backup & Site Recovery — Deployment & Configuration 13

Stage 1 — Assess . . . 13

Stage 2 — Configure (deploy hardened vaults + policies) . . . 14

Stage 3 — Pilot . . . 15

Stage 4 — Validate (prove the deployment works) . . . 16

Stage 5 — Expand . . . 16

5. Testing & Validation — Azure Backup & Site Recovery — Deployment & Configuration 17
Test 1 — Vault hardening . . . 17
Test 2 — Policy existence and assignment . . . 17
Test 3 — First successful recovery point . . . 18
Test 4 — Restore proof (into the isolated sandbox) . . . 18
Test 5 — ASR replication health and RPO . . . 18
Test 6 — Recovery-plan test failover . . . 18
Test 7 — Alerting . . . 19
Evidence to capture . . . 19
6. Troubleshooting — Azure Backup & Site Recovery — Deployment & Configuration 19
Backup — enablement and jobs . . . 19
Vault hardening . . . 20
Site Recovery — replication . . . 20
Recovery plans & test failover . . . 20
Alerting & reporting . . . 21
7. Best Practices — Azure Backup & Site Recovery — Deployment & Configuration 21
Vault hardening . . . 21
Policy design . . . 22
Site Recovery & recovery plans . . . 22
Operations . . . 22
Deferred decisions (customer-owned) . . . 22
Where to go next (adjacent catalog areas) . . . 23
8. Deliverables Summary — Azure Backup & Site Recovery — Deployment & Configuration 23
Documentation . . . 23
Automation scripts . . . 24
Named objects created (reference) . . . 24
Format . . . 24
Appendix A — PowerShell Scripts 25

Executive Summary — Azure Backup & Site Recovery — Deployment & Configuration

The problem

Hoping your VMs are backed up is not a disaster-recovery strategy. Many Azure estates run with no Recovery Services vault at all, with backup enabled on some VMs but not others, with policies that were set once and never revisited, and with **no** cross-region replication for the workloads that would actually take the business down if a region or a resource group were lost. When an accidental deletion, a ransomware event, or a regional outage happens, the team discovers — during the incident — that the data was never protected, that a "backup" was really just a mutable snapshot an attacker could delete, or that there is no tested way to bring the application back up somewhere else.

Two failure modes are especially common:

- **Backup exists but is not immutable or soft-delete-protected.** A backup an administrator (or

an attacker with administrator rights) can silently delete is not a ransomware control. Soft delete and immutability turn a backup vault into a last line of defense that survives credential compromise.

- **There is backup but no *recovery*.** Restoring a single file is not the same as failing an

application tier over to another region in sequence, with the database up before the app servers, DNS repointed, and post-failover scripts run. Without **Azure Site Recovery** replication and a tested **recovery plan**, "DR" is an aspiration, not a capability.

The solution

This solution **deploys and configures** Azure Backup and Azure Site Recovery end to end:

- **Recovery Services and Backup vaults** hardened with **soft delete** and **immutability**.
- **Backup policies** for **Azure VMs, Azure Files, SQL Server running in Azure VMs, and Azure

Blob storage**, assigned to the in-scope estate.

- **Azure Site Recovery** replication — **Azure-to-Azure** (region to region) and

on-premises-to-Azure — with a replication policy meeting the target RPO.

- **Recovery plans** with tiered sequencing, pre/post scripts, and a **non-disruptive test

failover**, plus** alerting **and a written** restore/failover runbook**.

It follows a consistent deployment model — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — so that protection is proven on a pilot before it is rolled out across the estate, and every object created is named to a convention and reused verbatim in the scripts and runbook.

> ***Scope boundary.*** *This solution *deploys and configures* Azure Backup and ASR. The ongoing discipline of *scheduling and evidencing test restores* across the estate is a separate program — see* ***Backup Validation & Recoverability Assurance*** *in this catalog. This solution proves the deployment works once (via test failover and a first restore); that program keeps proving it.*

What you get

• A hardened vault design with **soft delete + immutability** for Recovery Services and Backup vaults.

• **Backup policies** for VMs, Files, SQL-in-VM, and blobs, applied pilot-first then estate-wide.

• **ASR replication** (Azure-to-Azure and on-prem-to-Azure) with a policy meeting the agreed RPO.

• **Recovery plans** with sequencing, scripts, and a proven **test failover**.

• **Alerting** on backup/replication health and a **restore/failover runbook**.

• **Five automation artifacts** — a read-only discovery script, a Bicep template that provisions the hardened vaults and policies, a gated protection-enablement script, a read-only validation script (with an optional non-disruptive test failover), and a read-only reporting script.

Business outcomes

• **Provable protection** — workloads are backed up under policy, in immutable, soft-delete vaults, with defined RPO/RTO rather than assumed coverage.

• **Real recoverability** — replication plus a tested recovery plan means an application tier can actually be brought up in another region, in order, on demand.

• **Ransomware resilience** — immutability and soft delete keep recovery points intact even if an administrator account is compromised.

• **Operational readiness** — alerting and a written runbook mean the team executes a rehearsed procedure during an incident, not an improvised one.

> ***Your decision, not ours.*** *RPO/RTO targets are a business decision the customer owns — this solution implements the technical controls to *meet* the targets you set, and flags where a desired RPO/RTO is or is not achievable with the chosen configuration.*

Effort

A typical engagement is a **1–3 week** deployment: assessment and vault design in week 1, pilot configuration and a first test failover in week 2, and estate-wide expansion plus alerting and runbook handover in week 3. Delivered as a **Complete Solution Package** (PDF + Word + scripts).

1. Solution Overview — Azure Backup & Site Recovery — Deployment & Configuration

Description

This solution **deploys and configures** the two Azure services that together make workloads provably recoverable:

• **Azure Backup** — periodic, application-consistent recovery points for **Azure VMs**, **Azure

Files**,** SQL Server running inside Azure VMs**, and** Azure Blob storage**, stored in a** Recovery Services vault **(VM/Files/SQL) and a** Backup vault **(blobs and newer workloads), hardened with** soft delete **and** immutability**.

- **Azure Site Recovery (ASR)** — continuous **replication** of running workloads to a secondary

region (**Azure-to-Azure**) or from on-premises into Azure (**on-premises-to-Azure**), organized into **recovery plans** that fail workloads over in a defined **sequence**, run **pre/post scripts**, and can be exercised with a **non-disruptive test failover**.

The result is a backup and DR platform where every in-scope workload is protected under a named policy, recovery points cannot be silently destroyed, and a full application tier can be brought up in another region on demand — with the RPO/RTO the business agreed to.

How the technology works

- **Recovery Services vault** is the classic Azure Backup / ASR vault. It holds backup items for

Azure VMs, Azure Files, and SQL-in-VM, and it is also the vault type that ASR uses for replication. **Soft delete** retains deleted backup data for a recovery window (default 14 days, configurable up to 180) so deletions can be reversed; **immutability** (a vault-level setting, optionally **locked**) prevents recovery points from being deleted or shortened before their retention expires.

- **Backup vault** is the newer vault type used for **Azure Blob** backup, Azure Disks, and

PostgreSQL flexible server. It supports its own **soft delete** and **immutability** settings. Blob backup comes in two shapes: **operational** (continuous, local, point-in-time restore) and **vaulted** (periodic copies retained in the vault).

- **Backup policy** defines schedule (frequency), retention (daily/weekly/monthly/yearly points),

and — for SQL-in-VM — **log backup** frequency that sets the effective RPO. A policy is authored once and assigned to many items.

- **ASR replication** installs the **Mobility service** on each protected machine, continuously

ships changes to the target region/vault, and maintains recovery points per the **replication policy** (RPO target, recovery-point retention, app-consistent snapshot frequency).

- **Recovery plan** groups replicated items into ordered **groups** (e.g. Group 1 = database,

Group 2 = app, Group 3 = web) with **pre/post-action scripts or manual steps**, so a failover reconstitutes an application in the correct order. **Test failover** runs the plan into an isolated network **without** affecting the running production workload.

Deployment model — Assess → Configure → Pilot → Validate → Expand

Every solution in this catalog follows the same five-stage model so repeat buyers recognize the pattern:

1. **Assess** — inventory the estate (VMs, Files, SQL-in-VM, storage/blobs, on-prem servers),

determine what is protected today, and design the vault topology and target regions.

1. **Configure** — deploy the hardened vaults (soft delete + immutability) and the backup and

replication policies via Bicep, plan/what-if first.

1. **Pilot** — enable backup and replication on a small representative set, and run a **test failover** of a pilot recovery plan.

1. **Validate** — confirm vault hardening, policy coverage, replication health, first successful restore, and that the test failover met RTO; measure achieved RPO against target.

1. **Expand** — roll protection out across the estate, wire up alerting, and hand over the restore/failover runbook.

Named objects

This solution creates a defined set of objects. Each name is used **verbatim** in the implementation steps, the scripts, and the runbook. Adjust the placeholder region/scope tokens to your environment but keep the convention `<Purpose> – <Scope> – <Stage>`.

Object	Azure resource name (placeholder)	Purpose
Recovery Services Vault – Core – Configure	`rsv-backup-core-prod`	VM/Files/SQL backup + ASR
Backup Vault – Object – Configure	`bvault-backup-object-prod`	Blob (and disk) backup
VM Backup Policy – Daily-30d – Configure	`bkpol-vm-daily-30d`	Azure VM backup schedule/retention
Files Backup Policy – Daily-30d – Configure	`bkpol-files-daily-30d`	Azure Files backup
SQL Backup Policy – Log-15m – Configure	`bkpol-sql-log15m`	SQL-in-VM full/diff/log backup
Blob Backup Policy – Vaulted-30d – Configure	`bkpol-blob-vaulted-30d`	Vaulted blob backup
ASR Replication Policy – A2A-RPO – Configure	`asrpol-a2a-core`	Azure-to-Azure replication policy
Recovery Plan – Tier1 App – Pilot	`rp-tier1-app`	Ordered failover plan for the pilot app
DR Test Sandbox – Isolated – Validate	`rg-dr-test-sandbox`	Isolated RG/VNet for test failover & test restore
Backup & DR Alert Rule – Health – Expand	`alert-backupdr-health`	Alerts on backup/replication failures
Restore/Failover Runbook – Estate – Expand	*(document)*	The operational recovery procedure

Scope

In scope:

- Recovery Services vault and Backup vault deployment with soft delete + immutability.
- Backup policies and protection enablement for **Azure VMs, Azure Files, SQL-in-VM, and blobs**.
- ASR replication for **Azure-to-Azure** and **on-premises-to-Azure** (Hyper-V/VMware/physical via

the ASR configuration/process server or the modern appliance).

- Recovery plans with sequencing, scripts, and non-disruptive test failover.
- Backup/replication health **alerting** and a **restore/failover runbook**.
- One proving **test failover** and one proving **restore** to confirm the deployment works.

Out of scope (deferred or covered elsewhere):

- **Ongoing, scheduled test-restore evidence across the estate** — that is the

Backup Validation & Recoverability Assurance program in this catalog. This solution proves recoverability *once* at deployment; that program proves it *continuously*.

- **The RPO/RTO targets themselves** — those are a business decision the customer owns

(Requirement R1). This solution implements controls to meet them and flags feasibility.

- Microsoft 365 (Exchange/SharePoint/OneDrive/Teams) SaaS backup — that requires a third-party SaaS

backup product, not Azure Backup.

- Backup of workloads Azure Backup does not natively cover (e.g. some PaaS databases) beyond those

listed above — assess separately.

Authentication and connections used

- **Azure portal** (`portal.azure.com`) for interactive configuration.
- **Az PowerShell** (`Az.Accounts`, `Az.RecoveryServices`, `Az.DataProtection`, `Az.Compute`,

`Az.Storage`) for discovery, enablement, validation, and reporting.

- **Bicep / Azure Resource Manager** for declarative vault and policy deployment.
- **Azure RBAC** roles: **Backup Contributor** (configure backup), **Site Recovery Contributor**

(configure ASR), **Reader** (discovery/validation), and **Owner/Contributor** on the target subscription for the initial Bicep deployment. Grant the least role that completes each task.

2. Requirements — Azure Backup & Site Recovery — Deployment & Configuration

The category-level prerequisites are defined once in `global-requirements-backup-dr-solutions.md` (GR1–GR6: an administered subscription with backup tooling, a non-production restore target, documented RPO/RTO, secure evidence storage, change control, and a tooling host with connectivity). **Do not assume any of them are in place — verify each**

before starting. The requirements below are the **solution-specific** prerequisites that layer on top of the global set for deploying Azure Backup and Azure Site Recovery.

Each requirement is stated as **Description / Why required / How to verify / How to implement if missing.**

R1 — Agreed RPO/RTO targets per workload class (business decision)

Description: Written, stakeholder-agreed **Recovery Point Objective** and **Recovery Time Objective** per workload class (e.g. Tier-1 app VMs, file servers, SQL databases, blob data).

Why required: Backup frequency, log-backup interval, replication policy, and recovery-plan design are all derived from RPO/RTO. These targets are a **business decision the customer owns** — this solution implements controls to meet them, it does not set them.

How to verify: Locate an approved document stating RPO and RTO per workload class (this is GR3 applied to the specific workloads in scope here).

How to implement if missing: Facilitate a short workshop to agree targets per workload class, record them in the solution baseline, and note where a desired target is not achievable with the chosen tier of Azure Backup/ASR (flagged during the Configure stage).

R2 — A secondary (target) Azure region for ASR

Description: A chosen **target region** to replicate to (Azure-to-Azure), or a target region and landing resources for on-premises-to-Azure replication. The region must be a supported ASR pair for the source region and must have sufficient quota for the failed-over compute.

Why required: ASR replicates to a *different* region; failover has nowhere to go without target capacity. Insufficient vCPU/family quota in the target region causes failover to fail at the worst possible time.

How to verify: In the portal, check **Subscriptions > Usage + quotas** in the target region for the VM families you will fail over; confirm the source/target regions are a supported ASR pair (*verify current region-pair support in your tenant — pairings change*).

How to implement if missing: Select a supported target region, request quota increases for the required VM families, and confirm any required networking (target VNet/subnets) is planned.

R3 — Vault soft-delete and immutability policy decision

Description: A decision on the **soft-delete retention window** (14–180 days) and whether vault **immutability** is **enabled** or **enabled-and-locked** for each vault.

Why required: Soft delete and immutability are the ransomware controls that keep recovery points intact even under a compromised admin account. **Locked immutability is irreversible** — it cannot be disabled once locked, so the decision must be deliberate. This is a security/ops decision the customer's security function should own (defer per house rules).

How to verify: Confirm a documented decision exists specifying soft-delete days and immutability state (enabled vs. locked) per vault.

How to implement if missing: Present the tradeoff (stronger protection vs. irreversibility), obtain a signed decision, and apply it in the Bicep template (Section 3). Default recommendation: soft delete = 14 days (or longer per policy), immutability **enabled** (not locked) until the configuration has been validated, then locked deliberately.

R4 — RBAC roles for backup and Site Recovery

Description: The operating identity holds **Backup Contributor**, **Site Recovery Contributor**, and **Reader** as appropriate, and an identity with **Owner/Contributor** exists for the initial Bicep deployment.

Why required: Configuring backup, enabling replication, and creating vaults require distinct built-in roles. Using an over-privileged account for day-to-day operation violates least privilege; using an under-privileged one blocks the work.

How to verify: In **Subscriptions/Resource groups > Access control (IAM) > Role assignments**, confirm the operating identity's roles; run `Get-AzRoleAssignment` for the identity.

How to implement if missing: Assign **Backup Contributor** and **Site Recovery Contributor** at the appropriate scope for operations, and reserve an Owner/Contributor identity for the one-time infrastructure deployment.

R5 — Network line of sight for replication and (if applicable) the on-prem ASR appliance

Description: Outbound connectivity from source workloads/appliance to Azure ASR endpoints, and — for on-premises-to-Azure — a deployed **ASR replication appliance** (modern appliance) or the legacy configuration/process server, sized for the workload.

Why required: ASR ships change data continuously; blocked endpoints or an undersized appliance cause replication to fall behind RPO or fail to initialize.

How to verify: For Azure-to-Azure, confirm the Mobility service can reach ASR endpoints (no NSG/firewall block on required outbound). For on-prem, confirm the appliance is registered and healthy in the vault (**Site Recovery infrastructure**).

How to implement if missing: Allow the required ASR outbound URLs/service tags, and deploy and register the replication appliance per the target region and source hypervisor (Hyper-V / VMware / physical). *Verify current appliance requirements and supported source versions in your tenant.*

R6 — Application-consistency prerequisites for SQL-in-VM and app VMs

Description: The **SQL VSS writer** and the Azure VM **Guest Agent** are healthy on machines that need application-consistent recovery points; SQL-in-VM backup requires the workload registered via the vault's SQL backup extension.

Why required: Crash-consistent recovery points may not cleanly recover a transactional database. Application-consistent snapshots (VSS on Windows, pre/post scripts on Linux) are required for reliable SQL and app recovery.

How to verify: Confirm the VM Guest Agent shows **Ready**; for SQL-in-VM, confirm the databases are discoverable in the vault under **Backup > SQL Server in Azure VM**.

How to implement if missing: Install/repair the VM Guest Agent, ensure the VSS writer is healthy (`vssadmin list writers`), and register SQL-in-VM discovery in the vault before assigning the SQL backup policy.

R7 — An isolated DR test network and restore target (ties to GR2)

Description: An isolated resource group and virtual network — **DR Test Sandbox – Isolated – Validate** (`rg-dr-test-sandbox`) — with no routing to production, used for **test failover** and the proving **test restore**.

Why required: A test failover and a test restore must not collide with running production (duplicate IPs, name conflicts, or accidental data overwrite). Isolation is what makes proving safe.

How to verify: Confirm the isolated RG/VNet exists and is documented, and that it has no peering or routing to production networks.

How to implement if missing: Create the isolated RG and VNet (non-overlapping address space, no production peering) and record it as the standing test target.

R8 — Az PowerShell modules and Bicep tooling on the management host

Description: The tooling host (GR6) has current **Az** modules — `Az.Accounts`, `Az.RecoveryServices`, `Az.DataProtection`, `Az.Compute`, `Az.Storage` — and the **Bicep** CLI (via `az bicep` or the Azure CLI) installed.

Why required: The discovery, enablement, validation, and reporting scripts depend on these modules; the Bicep template requires the Bicep toolchain to deploy.

How to verify: Run `Get-Module Az.RecoveryServices,Az.DataProtection -ListAvailable` and `az bicep version`.

How to implement if missing: `Install-Module Az -Scope CurrentUser` (and the specific sub-modules above) and `az bicep install`. Confirm sign-in with `Connect-AzAccount`.

Global Requirements — Backup & DR Solutions

These requirements apply across **every** solution in the **Backup & DR Solutions** category. They are intentionally generic: any individual solution layers its specific prerequisites on top of these. **Do not assume any prerequisite is already in place** — for each, confirm **Description / Why / Verify / Implement if missing** before proceeding.

GR1 — A tenant/subscription with backup tooling and administrative access

Description: A cloud tenant and/or subscription (for example a Microsoft 365 tenant and/or an Azure subscription) with backup tooling provisioned, plus administrative access sufficient to view, configure, and

restore protected data.

Why: Backup and DR work — coverage assessment, restore testing, remediation — is impossible without administrative reach into the platform that holds the data and the tooling that protects it.

Verify: Sign in to the relevant admin console/portal and confirm an account with the required administrative or backup-operator role, and that backup tooling is present and licensed.

Implement if missing: Provision the tenant/subscription, license and configure the backup tooling, and obtain an appropriately scoped administrative/operator role before continuing.

GR2 — A non-production restore target

Description: An isolated, non-production destination for performing test restores — a sandbox location, alternate path, or restore-test resource group/network — kept separate from live data and systems.

Why: Recovery testing must prove restorability **without** risking production. Restoring into production can overwrite live data or cause resource collisions and outages.

Verify: Confirm a documented, isolated restore target exists for each in-scope workload type and that it cannot affect production data or networks.

Implement if missing: Provision a dedicated non-production restore target (sandbox location / alternate path / restore-test resource group with an isolated network) and document it.

GR3 — Documented RPO/RTO objectives

Description: Written, agreed **Recovery Point Objective** and **Recovery Time Objective** targets per workload class.

Why: Backup and recovery outcomes can only be judged PASS/FAIL against defined objectives. Without them, "recoverable" and "timely" are subjective and unauditable.

Verify: Locate an approved document stating RPO and RTO per workload class.

Implement if missing: Propose objectives per workload class, agree them with the stakeholder/client, and record them in the solution's baseline.

GR4 — Secure evidence storage

Description: A controlled, retained store for recovery-test evidence and reporting outputs, with access control and a defined retention period.

Why: Audits, insurers, and clients require durable proof of tested recovery. Evidence that is ad hoc, mutable, or short-lived is not defensible.

Verify: Confirm a designated store exists with access control and retention, holding prior evidence/reports.

Implement if missing: Designate a secure, access-controlled store with a documented retention period and restrict write access to program operators.

GR5 — Change control

Description: A change-management process under which backup/restore/DR activities (test restores, on-demand backups, configuration changes) are authorized and logged.

Why: These activities consume resources, create artifacts, and touch protected data; they must be authorized and traceable, especially in shared or regulated environments.

Verify: Confirm a change record or standing change covers the recurring backup/DR activities.

Implement if missing: Establish a standing/recurring change authorizing the activity and require a change reference on each execution.

GR6 — Network connectivity and a tooling host

Description: A management host with the required administrative modules/CLIs installed (for example Az PowerShell and the Microsoft Graph PowerShell SDK) and network access to the relevant portals, APIs, and backup consoles.

Why: Discovery, validation, reporting, and remediation tooling — and the restores themselves — require reliable connectivity and the right client tools.

Verify: From the host, confirm the required modules/CLIs are installed and the relevant endpoints are reachable.

Implement if missing: Install the required tooling and ensure outbound connectivity to the platform and backup console from the host.

3. Implementation — Azure Backup & Site Recovery — Deployment & Configuration

This section deploys the platform in five stages: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**. Steps are imperative and ordered. Portal navigation paths are given in full. Every object created uses its verbatim name from the **Named objects** table in Section 1 (e.g. **Recovery Services Vault – Core – Configure** = `rsv-backup-core-prod`). Placeholder values (`contoso`, region names, subscription IDs) are examples — substitute your own.

> *Complete the requirements (Section 2 and the global requirements) first. Do not assume any prerequisite; verify each before starting. Anything disruptive is done **pilot-first** and, for scripts, `-WhatIf` **/ plan first**.*

Stage 1 — Assess

Goal: know what exists, what is protected, and what the target topology will be.

1. Sign in to the management host and connect: `Connect-AzAccount`, then

`Set-AzContext -SubscriptionId <sub-id>`.

1. Run `BCKDR-Discovery.ps1` (read-only). It enumerates VMs, Azure Files shares, SQL-in-VM

candidates, storage accounts/blob containers, existing Recovery Services and Backup vaults, and any existing ASR replication, and flags each in-scope resource as **protected** or **unprotected**. Save its CSV as the assessment baseline.

1. From the CSV, build the protection plan: for each workload class record the source region,

intended vault, intended policy, and (for DR-tier workloads) whether it needs **ASR replication** to the target region (R2).

1. Confirm the design decisions from Section 2 are recorded: RPO/RTO per class (R1), target region

(R2), soft-delete window and immutability state (R3), and the isolated **DR Test Sandbox – Isolated – Validate** (`rg-dr-test-sandbox`, R7).

Stage 2 — Configure (deploy hardened vaults + policies)

Goal: deploy the vaults and policies declaratively, hardened with soft delete and immutability.

1. Review `BCKDR-Deploy.bicep`. It creates, in the primary region:

• **Recovery Services Vault – Core – Configure** (`rsv-backup-core-prod`) with **soft delete

enabled **and** immutability** set per R3.

• **Backup Vault – Object – Configure** (`bvault-backup-object-prod`) with its own soft delete

and immutability.

• Backup policies: **VM Backup Policy – Daily-30d – Configure** (`bkpol-vm-daily-30d`),

Files Backup Policy – Daily-30d – Configure (`bkpol-files-daily-30d`), **SQL Backup Policy – Log-15m – Configure** (`bkpol-sql-log15m`), and **Blob Backup Policy – Vaulted-30d – Configure** (`bkpol-blob-vaulted-30d`).

1. Set parameters (region, soft-delete days, immutability state, retention) in a

`BCKDR-Deploy.parameters.json` or inline, matching the R1/R3 decisions.

1. **Preview before applying** — run a what-if:

```
az deployment group what-if --resource-group rg-backup-core-prod \
  --template-file BCKDR-Deploy.bicep --parameters @BCKDR-Deploy.parameters.json
```

Review the planned changes. Only then deploy:

```
az deployment group create --resource-group rg-backup-core-prod \
  --template-file BCKDR-Deploy.bicep --parameters @BCKDR-Deploy.parameters.json
```

1. Confirm hardening in the portal: **portal.azure.com > Recovery Services vaults >

`rsv-backup-core-prod` > Properties**. Under** Security Settings / Soft Delete **confirm soft delete is** Enabled**; under** Immutable vault **confirm the chosen state. Do the same for** Backup vaults > `bvault-backup-object-prod` > Properties**.

> *Leave immutability **enabled but not locked** until after Validate (Stage 4). Locking is **irreversible** — lock deliberately in Stage 5 once the configuration is proven.*

1. Create the **ASR Replication Policy – A2A-RPO – Configure** (`asrpol-a2a-core`): in

`rsv-backup-core-prod` **> Site Recovery infrastructure > Replication policies > + Replication policy**, set the **RPO threshold**, **recovery-point retention**, and **app-consistent snapshot frequency** to meet the agreed RPO (R1). (The scripts reference this policy by name.)

Stage 3 — Pilot

Goal: protect a small, representative set and prove failover before touching the whole estate. Pick 1–2 VMs per workload class (a file server, a SQL-in-VM, a Tier-1 app VM) as the pilot.

3a — Enable Azure Backup on pilot workloads

1. Preview enablement with `BCKDR-EnableProtection.ps1` — it defaults to `-WhatIf` and only *reports* what it would protect:

```
.\BCKDR-EnableProtection.ps1 -Vault rsv-backup-core-prod -Scope Pilot -WhatIf
```

1. Apply to the pilot only, gated behind change control (GR5):

```
.\BCKDR-EnableProtection.ps1 -Vault rsv-backup-core-prod -Scope Pilot -Apply -ChangeRef <CR>
```

This assigns **VM Backup Policy – Daily-30d – Configure** to pilot VMs, **Files Backup Policy – Daily-30d – Configure** to the pilot file share, and registers the pilot SQL-in-VM databases under **SQL Backup Policy – Log-15m – Configure**.

1. Alternatively, enable manually in the portal: `rsv-backup-core-prod` **> Backup > + Backup >** choose the datasource type, select the policy by name, and select the pilot items.

1. Trigger a first backup: on each pilot item, **Backup items > (item) > Backup now**. Wait for the first recovery point to complete.

3b — Enable ASR replication on a pilot app (Azure-to-Azure)

1. In `rsv-backup-core-prod` **> Site Recovery > Enable Site Recovery > Azure virtual machines**, choose the source region, the pilot VM(s), the **target region** (R2), the target resource group and VNet, and the **ASR Replication Policy – A2A-RPO – Configure** (`asrpol-a2a-core`).

1. Start replication and wait until the pilot VM(s) reach **Protected** with an initial recovery point (initial replication can take time proportional to disk size).

> *For **on-premises-to-Azure**, first register the ASR replication appliance (R5) under **Site Recovery infrastructure**, then **Enable replication** selecting the on-prem source machines, target region, and policy. *Verify current appliance and source-OS support in your tenant.**

3c — Build the pilot recovery plan

1. In **`rsv-backup-core-prod` > Site Recovery > Recovery Plans (Site Recovery) > + Recovery plan**, **create** Recovery Plan – Tier1 App – Pilot** (`rp-tier1-app`). Add the replicated pilot items.

1. **Customize** the plan into ordered **groups** to reflect dependency order — e.g. **Group 1** = database VM, **Group 2** = application VM, **Group 3** = web VM.

1. Add **pre/post actions** where needed: a post-Group-1 step to confirm the database is accepting

connections, and a post-failover manual/scripted step to repoint DNS. Keep scripts idempotent.

Stage 4 — Validate (prove the deployment works)

Goal: prove — once — that backup restores and the recovery plan fails over within target. (Ongoing scheduled restore-testing across the estate is the separate **Backup Validation & Recoverability Assurance** program; this stage is the one-time deployment proof.)

1. Run `BCKDR-Validate.ps1` (read-only). It confirms: soft delete + immutability are set on both

vaults, the four policies exist and are assigned to the pilot items, pilot items have a recent successful recovery point, and ASR pilot items are **Protected** with an RPO within target. Save its PASS/FAIL output as deployment evidence.

1. **Prove restore** — perform one file-level restore from a pilot VM recovery point into the

DR Test Sandbox – Isolated – Validate (`rg-dr-test-sandbox`): **Backup items > (pilot VM) > File Recovery**, mount the recovery point, and confirm a known file is present and readable. Do **not** restore over production.

1. **Prove failover** — run a **non-disruptive test failover** of **Recovery Plan – Tier1 App –

Pilot** into the isolated network. Either:

• Portal: **Recovery Plans >** `rp-tier1-app` **> Test Failover**, select the latest recovery point

and the `rg-dr-test-sandbox` VNet; or

• Script (gated): `.\BCKDR-Validate.ps1 -RunTestFailover -RecoveryPlan rp-tier1-app

-TestNetworkResourceGroup rg-dr-test-sandbox -ChangeRef <CR>`.

Test failover does **not** affect the running production workload. Confirm the failed-over VMs boot in order in the isolated network and the app responds. Record the elapsed time and compare it to the **RTO** target (R1).

1. **Clean up the test failover** — always run **Cleanup test failover** on the recovery plan

afterward (portal: **Recovery Plans >** `rp-tier1-app` **> Cleanup test failover**) to delete the test VMs and release resources. Delete any test-restore artifacts from `rg-dr-test-sandbox`.

1. If restore, test failover, or RPO/RTO checks fail, resolve before expanding (see Section 6).

Stage 5 — Expand

Goal: roll protection across the estate, add alerting, lock hardening, and hand over the runbook.

1. **Expand backup** — re-run `BCKDR-EnableProtection.ps1` with `-Scope Estate` in `-WhatIf`

first, review, then `-Apply -ChangeRef <CR>` to assign the named policies across all remaining in-scope VMs, file shares, SQL-in-VM databases, and blob accounts.

1. **Expand replication** — enable ASR on the remaining DR-tier workloads against

`asrpol-a2a-core`, and extend or add recovery plans (following the `rp-<tier>-<app>` naming) with correct sequencing per application.

1. **Configure alerting** — create **Backup & DR Alert Rule – Health – Expand**

(`alert-backupdr-health`): in `rsv-backup-core-prod` **> Monitoring > Backup Alerts** (and Azure Monitor alert rules) enable alerts for **backup failures**, **restore failures**, and **ASR replication health / RPO breaches**, routed to an action group (email/Teams/ITSM). Confirm at least one test alert reaches the destination.

1. **Lock immutability (deliberate, irreversible)** — now that the configuration is validated, if

R3 calls for locked immutability, set both vaults' immutability to **Locked** in **Vault > Properties > Immutable vault**. Understand this cannot be undone.

1. **Hand over the runbook** — finalize the **Restore/Failover Runbook – Estate – Expand**

document: how to run a real failover of each recovery plan (and **failback**), how to restore each workload type, the alert-response steps, the RPO/RTO targets, and who authorizes a failover. Cross-reference the **Backup Validation & Recoverability Assurance** program for the ongoing restore-testing cadence.

1. Run `BCKDR-Report.ps1` (read-only) to produce the estate-wide posture report (protected vs.

unprotected, policy coverage, replication health, last recovery point per item) for the handover pack.

5. Testing & Validation — Azure Backup & Site Recovery — Deployment & Configuration

Testing confirms three things: the **vaults are hardened**, the **backup policies actually protect the estate**, and **recovery works** (restore + failover) within target. Use the scripts by name; each is read-only unless explicitly gated. Record results as deployment evidence in the secure store (GR4).

> ***Boundary.*** *This section validates the *deployment* (does it work, once). The ongoing, scheduled discipline of proving restores across the estate over time is the separate* ***Backup Validation & Recoverability Assurance*** *program — do not duplicate that cadence here.*

Test 1 — Vault hardening

Procedure: Run `BCKDR-Validate.ps1` and check the portal (**Recovery Services vaults >** `rsv-backup-core-prod` **> Properties**, and **Backup vaults >** `bvault-backup-object-prod` **> Properties**).

Pass criteria:

- **Soft delete** is **Enabled** on both vaults, with the retention window from R3.
- **Immutability** is in the state agreed in R3 (enabled during pilot; locked only after Stage 4).
- Vaults exist in the intended region and resource group.

Test 2 — Policy existence and assignment

Procedure: Run `BCKDR-Validate.ps1`; cross-check `rsv-backup-core-prod` **> Backup items**.

Pass criteria:

• All four policies exist: **VM Backup Policy – Daily-30d – Configure**, **Files Backup Policy – Daily-30d – Configure**, SQL Backup Policy – Log-15m – Configure**,** Blob Backup Policy – Vaulted-30d – Configure**.

• Every in-scope pilot (then, post-Expand, estate) item is assigned to the correct policy — **no in-scope item is unprotected**.

• Schedule/retention on each policy matches the RPO/RTO decisions (R1). For SQL-in-VM, the **log backup interval** (e.g. 15 min) matches the SQL RPO.

Test 3 — First successful recovery point

Procedure: For each pilot item, **Backup items > (item) > Last backup status**; or read `BCKDR-Validate.ps1` output.

Pass criteria:

• Each pilot item shows a **successful** recent recovery point (data age within the RPO).

• No item is stuck in **Warning** / initial-backup-never-completed.

Test 4 — Restore proof (into the isolated sandbox)

Procedure: Perform one **File Recovery** restore from a pilot VM recovery point into **DR Test Sandbox – Isolated – Validate** (`rg-dr-test-sandbox`). Confirm a known file is present and readable. **Never restore over production.**

Pass criteria:

• The restore completes within the workload's **RTO**.

• The restored file is present and its content is intact.

• Nothing in production was modified.

Test 5 — ASR replication health and RPO

Procedure: Run `BCKDR-Validate.ps1`; check `rsv-backup-core-prod` **> Site Recovery > Replicated items**.

Pass criteria:

• Pilot (then estate) DR-tier items are **Protected** with a healthy replication status.

• Current **RPO** is at or below the target set in **ASR Replication Policy – A2A-RPO – Configure**.

• No replication errors or "RPO exceeded" alerts are open.

Test 6 — Recovery-plan test failover

Procedure: Run a **non-disruptive test failover** of **Recovery Plan – Tier1 App – Pilot** (`rp-tier1-app`) into `rg-dr-test-sandbox` (portal **Test Failover**, or `BCKDR-Validate.ps1 -RunTestFailover`). **Always run Cleanup test failover afterward.**

Pass criteria:

- Groups fail over in the defined order (e.g. DB → app → web); pre/post actions run successfully.
- Failed-over VMs boot in the isolated network and the application responds.
- Total elapsed time is within the application's **RTO**.
- Production is unaffected throughout; **Cleanup test failover** removes all test resources.

Test 7 — Alerting

Procedure: Confirm **Backup & DR Alert Rule – Health – Expand** (`alert-backupdr-health`) is configured and fires. Trigger a benign test (e.g. an alert-rule test notification or a controlled failed-job condition in a lab).

Pass criteria:

- Backup-failure, restore-failure, and ASR-health/RPO-breach alerts are enabled.
- At least one test alert reaches the action group destination (email/Teams/ITSM).

Evidence to capture

For each test, record: date/time, operator, the object name(s) involved, the recovery point used, elapsed time vs. RTO, data age vs. RPO, PASS/FAIL, and the change reference (GR5). Store in the secure evidence store (GR4). Run `BCKDR-Report.ps1` to compile the estate-wide posture summary for the handover pack. Any FAIL routes to Section 6 (Troubleshooting) before go-live.

6. Troubleshooting — Azure Backup & Site Recovery — Deployment & Configuration

Common failures, in **symptom** → **cause** → **fix** form. Most map back to a requirement (Section 2) or a stage in the implementation (Section 3).

Backup — enablement and jobs

Symptom: `Enable-AzRecoveryServicesBackupProtection` or portal enablement fails with authorization error. **Cause:** The operating identity lacks **Backup Contributor** at the vault scope (R4). **Fix:** Assign **Backup Contributor** on the vault/resource group, sign out/in, and retry. `BCKDR-EnableProtection.ps1` will preview first in `-WhatIf` — confirm the plan, then `-Apply`.

Symptom: First VM backup stays in **Warning** or never completes. **Cause:** VM **Guest Agent** not ready, or an extension/snapshot failure (R6). **Fix:** Confirm the VM Guest Agent shows **Ready**; restart the VM Agent; check the VM's **Backup > Backup jobs** for the sub-error. For Windows, confirm the VSS writers are

healthy (`vssadmin list writers`). Re-run **Backup now**.

Symptom: SQL-in-VM databases don't appear to protect. **Cause:** The SQL workload isn't **registered/discovered** in the vault, or the AzureWLBackup extension is unhealthy (R6). **Fix:** In `rsv-backup-core-prod` **> Backup > SQL Server in Azure VM**, run **Discover DBs**, confirm the VM registers, then assign **SQL Backup Policy – Log-15m – Configure**. Verify the log backup interval matches your SQL RPO.

Symptom: Azure Files backup enablement fails or the share is not listed. **Cause:** The storage account isn't reachable by the vault, or the share is in an unsupported account/redundancy tier. **Fix:** Confirm the storage account is in a supported configuration and region; register the storage account under **Backup > Azure Files**, then assign **Files Backup Policy – Daily-30d – Configure**.

Vault hardening

Symptom: Bicep deployment fails on `securitySettings` / `immutabilitySettings`. **Cause:** The API version in `BCKDR-Deploy.bicep` doesn't match the property shape your tenant expects — these paths drift between API versions. **Fix:** Run `az deployment group what-if` to see the rejected property, and adjust the `@2024-04-01` API version / property path to the current one. *Verify current property support in your tenant.*

Symptom: You cannot disable immutability or shorten retention. **Cause:** Immutability was set to **Locked** — this is **irreversible** by design (R3). **Fix:** There is no fix — locked immutability cannot be reversed. This is why the implementation keeps immutability **Unlocked** until Stage 4 (Validate) and only locks in Stage 5. If you need a different configuration, it must be a new vault.

Symptom: A recovery point you deleted is gone sooner than expected. **Cause:** Soft delete disabled, or its window shorter than assumed. **Fix:** Confirm **soft delete Enabled** with the intended retention (R3) on both vaults; recover in-window items from **soft-deleted** state; extend the window going forward.

Site Recovery — replication

Symptom: ASR replicated item stuck in initial replication or **Critical** health. **Cause:** Blocked outbound to ASR endpoints, or an unhealthy Mobility service / appliance (R5). **Fix:** Allow the required ASR outbound URLs/service tags on NSGs/firewall; for on-prem, confirm the replication appliance is **registered and healthy** under **Site Recovery infrastructure**; reinstall/repair the Mobility service on the source machine.

Symptom: Current **RPO exceeds** the target. **Cause:** Insufficient bandwidth, high change rate, or an undersized appliance/policy (R5). **Fix:** Check replication health and the churn rate; increase bandwidth or scale the appliance; re-evaluate whether the target RPO is achievable and, if not, surface it — RPO is a business decision (R1) that must match reality.

Symptom: `Enable replication` cannot select a target because of quota. **Cause:** The target region lacks vCPU/family quota for the failed-over compute (R2). **Fix:** Request a quota increase for the required VM families in the target region before enabling replication or failing over.

Recovery plans & test failover

Symptom: Test failover boots VMs but the application doesn't come up. **Cause:** Recovery-plan groups are not sequenced by dependency (app started before its database). **Fix:** Edit **Recovery Plan – Tier1 App – Pilot** so **Group 1** = database, **Group 2** = app, **Group 3** = web, and add a post-Group-1 check that the DB is accepting connections before later groups start.

Symptom: Test failover succeeds but leaves resources/cost behind. **Cause: Cleanup test failover** was not run. **Fix:** Always run **Recovery Plans >** `rp-tier1-app` **> Cleanup test failover** after validating. Delete any leftover artifacts in **DR Test Sandbox – Isolated – Validate** (`rg-dr-test-sandbox`).

Symptom: Test failover VMs land on a production network / cause IP conflicts. **Cause:** The wrong target network was selected. **Fix:** Always run test failover into the isolated `rg-dr-test-sandbox` VNet (R7). Never point a test failover at a production network.

Symptom: `BCKDR-Validate.ps1 -RunTestFailover` throws before starting. **Cause:** The gate — it requires `-RecoveryPlan`, `-TestNetworkResourceGroup`, and `-ChangeRef`. **Fix:** Supply all three; the gate is intentional so a test failover is never triggered casually.

Alerting & reporting

Symptom: Backups fail silently; nobody is notified. **Cause: Backup & DR Alert Rule – Health – Expand** (`alert-backupdr-health`) not configured or not routed to a monitored action group. **Fix:** Enable backup-failure, restore-failure, and ASR RPO-breach alerts and route them to an action group; send a test notification and confirm it arrives.

Symptom: `BCKDR-Discovery.ps1` / `BCKDR-Report.ps1` returns nothing. **Cause:** Wrong subscription context or missing **Reader** access. **Fix:** `Set-AzContext -SubscriptionId <id>`; confirm Reader on the subscription and vaults; re-run. Newly created resources may not appear until the first run after they exist — re-baseline on a cadence.

7. Best Practices — Azure Backup & Site Recovery — Deployment & Configuration

Keep the platform clean, defensible, and cheap to run.

Vault hardening

- **Enable soft delete on every vault, always.** It is the free undo button for accidental or malicious deletion. Set the retention window per your data-protection policy (14–180 days).
- **Enable immutability, and lock it only after validation.** Locked immutability is **irreversible** — prove the whole configuration first (Stage 4), then lock deliberately. Locking before validation can trap a mistake you cannot undo.
- **Use multi-user authorization (MUA) for critical vault operations** where available, so a single compromised admin cannot weaken protection. *Verify current MUA availability in your tenant.*
- **Separate the backup blast radius.** Keep vaults in a dedicated, tightly-RBAC'd resource group; restrict **Backup Contributor** / **Site Recovery Contributor** to the people who operate them.

Policy design

• **Author policies once, assign many.** Reuse the named policies (`bkpol-vm-daily-30d`, etc.) rather than creating per-VM policies — it keeps RPO/RTO consistent and auditable.

• **Match policy to RPO/RTO, not to habit.** For SQL-in-VM, the **log-backup interval** *is* your RPO — set it to the agreed target (R1), not a default. Retention (daily/weekly/monthly/yearly) should reflect real recovery and compliance needs, not "keep everything forever" (cost) or "keep 7 days" (too little).

• **Right-size retention for cost.** Long-term retention points accrue storage cost; use tiered retention (short daily + longer monthly/yearly) instead of long daily retention.

Site Recovery & recovery plans

• **Sequence recovery plans by dependency.** Database group first, then app, then web. A plan that boots everything at once often fails app health checks.

• **Keep pre/post scripts idempotent and version-controlled.** They run under stress during a real failover; they must be safe to re-run.

• **Test failover regularly and always clean up.** A test failover proves the plan; leaving test resources behind wastes money and clutters the target region. Automate the cleanup step.

• **Plan and rehearse failback.** Failover is half the story — document and test the return to the primary region so DR is not a one-way trip.

• **Watch target-region quota.** Failover fails if the target region lacks vCPU/family quota (R2) — recheck quota as the estate grows.

Operations

• **Alert on failure, not just success.** Route backup-failure, restore-failure, and ASR RPO-breach alerts to a monitored destination (`alert-backupdr-health`); a silent backup failure is the gap you find during an incident.

• **Keep the runbook current.** Update **Restore/Failover Runbook – Estate – Expand** whenever the estate, RPO/RTO, or recovery plans change.

• **Re-baseline coverage as the estate grows.** New VMs, shares, and databases drift out of protection unless re-assessed — re-run `BCKDR-Discovery.ps1` on a cadence.

• **Never hand-edit production during a test.** All proving happens in `rg-dr-test-sandbox`.

Deferred decisions (customer-owned)

• **RPO/RTO targets** are a business decision (R1) — this solution meets them, it does not set them.

• **Immutability lock and soft-delete window** are security/ops decisions (R3) — obtain sign-off.

• **Retention beyond operational need** may be driven by legal/records requirements — route those to the customer's compliance function, not the backup admin.

Where to go next (adjacent catalog areas)

• **Backup Validation & Recoverability Assurance** — the essential companion: a scheduled, evidence-producing discipline that proves restores across the estate *continuously* (this solution proves recoverability once, at deployment).

• **Backup & DR Solutions** category — broader resilience patterns (immutable/offsite copies, ransomware recovery runbooks) that build on this deployment.

• **Azure governance / cost management** — policy-as-code to *enforce* that every new VM lands under a backup policy, and cost controls for backup/replication storage.

8. Deliverables Summary — Azure Backup & Site Recovery — Deployment & Configuration

Documentation

Document	Purpose
`executive-summary.md`	The problem, the solution, outcomes, scope boundary, and effort.
`01-solution-overview.md`	What it is, how Azure Backup + ASR work, the Assess→Configure→Pilot→Validate→Expand model, named objects, scope, and connections.
`02-requirements.md`	Solution-specific prerequisites (R1–R8) layered on the category global requirements.
`03-implementation.md`	Ordered, click-by-click deployment across the five stages, naming every object.
`05-testing.md`	Seven validation tests (vault hardening, policy coverage, restore proof, replication/RPO, test failover, alerting) with pass criteria.
`06-troubleshooting.md`	Symptom → cause → fix for the common Azure Backup and ASR failures.
`07-best-practices.md`	Hardening, policy, ASR, and operations guidance, plus where to go next.
`08-deliverables-summary.md`	This summary.
`global-requirements-backup-dr-solutions.md`	Category-wide prerequisites (GR1–GR6).

Automation scripts

Script	Purpose
`BCKDR-Discovery.ps1`	**Read-only.** Inventories VMs, Azure Files, SQL-in-VM, storage/blobs, existing vaults, and ASR; flags each in-scope resource as protected or unprotected to build the assessment baseline.
`BCKDR-Deploy.bicep`	**Plan/what-if first.** Declarative deployment of the Recovery Services vault and Backup vault (soft delete + immutability) and the four backup policies.
`BCKDR-EnableProtection.ps1`	`-WhatIf` **by default,** `-Apply` **gated.** Assigns the named backup policies and enables protection on pilot then estate VMs/Files/SQL-in-VM/blobs under change control.
`BCKDR-Validate.ps1`	**Read-only** (optional gated `-RunTestFailover`). Verifies vault hardening, policy assignment, recovery-point recency, and ASR replication health/RPO; can trigger a non-disruptive test failover of a recovery plan.
`BCKDR-Report.ps1`	**Read-only.** Compiles estate-wide backup/DR posture (protected vs. unprotected, policy coverage, replication health, last recovery point) into a CSV for the handover pack.

Named objects created (reference)

`rsv-backup-core-prod` · `bvault-backup-object-prod` · `bkpol-vm-daily-30d` · `bkpol-files-daily-30d` · `bkpol-sql-log15m` · `bkpol-blob-vaulted-30d` · `asrpol-a2a-core` · `rp-tier1-app` · `rg-dr-test-sandbox` · `alert-backupdr-health` · Restore/Failover Runbook.

Format

This solution is delivered as a **Complete Solution Package (PDF + Word + scripts)**: the eight documentation files compiled to PDF and Word, plus the five automation artifacts, ready to execute against an Azure subscription with the roles in Section 2.

> ***Companion solution.*** *Pair this deployment with* ***Backup Validation & Recoverability Assurance*** *for the ongoing, scheduled restore-testing and evidence discipline. This solution *builds and proves* the platform; that program *keeps proving* it.*

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

BCKDR-Deploy.bicep

Purpose: Declarative deployment for the "Azure Backup & Site Recovery - Deployment & Configuration" solution (Stage 2 - Configure). Creates: - Recovery Services Vault - Core - Configure (rsv-backup-core-prod) with SOFT DELETE enabled and IMMUTABILITY per parameter. - Backup Vault - Object - Configure (bvault-backup-object-prod) with its own soft delete + immutability, for blob backup. - VM Backup Policy - Daily-30d - Configure (bkpol-vm-daily-30d) - Files Backup Policy - Daily-30d - Configure (bkpol-files-daily-30d) (SQL-in-VM and vaulted-blob policies are created in-console/CLI after the SQL workload and blob datasource are registered - see 03-implementation.md.)

BCKDR-Discovery.ps1

Purpose: Read-only assessment for the "Azure Backup & Site Recovery - Deployment & Configuration" solution. Enumerates Azure VMs, Azure Files shares, SQL-in-VM candidates, storage accounts, and any existing Recovery Services vaults, Backup vaults, and ASR replicated items. Flags each in-scope resource as PROTECTED or UNPROTECTED to build the assessment baseline (Stage 1 - Assess).

How to run: Connect-AzAccount Set-AzContext -SubscriptionId <sub-id> .\BCKDR-Discovery.ps1 .\BCKDR-Discovery.ps1 -SubscriptionId <sub-id> -OutputPath .\baseline.csv

BCKDR-EnableProtection.ps1

Purpose: Enablement helper for the "Azure Backup & Site Recovery - Deployment & Configuration" solution (Stage 3 - Pilot / Stage 5 - Expand). Assigns the named backup policies (bkpol-vm-daily-30d, bkpol-files-daily-30d) to in-scope resources and enables Azure Backup protection.

How to run: Preview (no changes): .\BCKDR-EnableProtection.ps1 -Vault rsv-backup-core-prod -Scope Pilot -WhatIf Apply to pilot, gated: .\BCKDR-EnableProtection.ps1 -Vault rsv-backup-core-prod -Scope Pilot -Apply -ChangeRef CR-1234 Later, expand to the estate: .\BCKDR-EnableProtection.ps1 -Vault rsv-backup-core-prod -Scope Estate -Apply -ChangeRef CR-1234

BCKDR-Report.ps1

Purpose: Read-only reporting for the "Azure Backup & Site Recovery - Deployment & Configuration" solution (Stage 5 - Expand / handover). Produces a single CSV summarizing, per subscription: protected vs. unprotected resources, which policy protects each item, last recovery point and status, ASR replication health, and vault hardening state - suitable for the handover pack and periodic posture review.

How to run: Connect-AzAccount .\BCKDR-Report.ps1 .\BCKDR-Report.ps1 -SubscriptionId <sub-id> -OutputPath .\posture.csv

BCKDR-Validate.ps1

Purpose: Read-only validation for the "Azure Backup & Site Recovery - Deployment & Configuration" solution (Stage 4 - Validate). Checks, per Section 5: - Soft delete + immutability set on the Recovery Services and Backup vaults. - The named backup policies exist and are assigned (no in-scope item unprotected). - Protected items have a recent successful recovery point. - ASR replicated items are

Protected and within RPO. Emits a PASS/FAIL line per check as deployment evidence.

How to run: .\BCKDR-Validate.ps1 -Vault rsv-backup-core-prod -BackupVault bvault-backup-object-prod .\BCKDR-Validate.ps1 -Vault rsv-backup-core-prod -RunTestFailover ` -RecoveryPlan rp-tier1-app -TestNetworkResourceGroup rg-dr-test-sandbox -ChangeRef CR-1234